Block practice · Series and the binomial expansion

Work through each question type in order. Each section has a worked-example retry and all further practice. Attempt only the part outlined in red.

Expanding a bracket with a negative or fractional power

When is a binomial expansion valid?

Getting a bracket ready for the binomial series: a(1 + bx)^n

Finding the first term and common difference of an arithmetic series

Using a binomial expansion to estimate a value

Finding n from an equation linking sums and terms

Finding the first term and common ratio of a geometric series

Does a geometric series have a sum to infinity?

Sum to infinity of a geometric series

Adding up a finite geometric series

Finding n in a geometric series with logarithms

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Practice workbook · Expanding a bracket with a negative or fractional power

Expanding (1 + kx)^n for rational n in ascending powers of x up to a stated term, with coefficients simplified.

18 question parts: 1 worked-example retry and 17 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-SER-18

1. Question 9 · part a · 3 marks

2020 JAN · Paper 01

Worked-example retry · Attempt part a only.

9 (a) Expand $\sqrt{1-4x}$ in ascending powers of x up to and including the term in x^3, giving each coefficient as an integer.

(3)

(b) Use your expansion with a suitable value for x to obtain an estimate of $\sqrt{0.76}$
Give your answer correct to 4 decimal places.

(3)

(c) Hence find, to 3 decimal places, an estimate of $\sqrt{19}$

(2)

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1. Question 9 · part a · 3 marks · Working and answer

Working and answer

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2. Question 9 · part a · 3 marks

2022 JAN · Paper 01

Further practice · Attempt part a only.

9 (a) Expand $\frac{1}{\sqrt{1-2x}}$ in ascending powers of x up to and including the term in x^3

Simplify each term as far as possible.

(3)

(b) Show that $\frac{1}{\sqrt{0.96}} = \frac{5\sqrt{6}}{12}$

Show all your working clearly.

(2)

(c) Express $\frac{1}{5\sqrt{6}-12}$ in the form $\frac{a\sqrt{6}}{c} + b$ where a, b and c are integers.

Show all your working clearly.

(2)

Using the expansion you found in part (a) with a suitable value of x

(d) find an estimate, to 5 decimal places, of $\frac{9}{5\sqrt{6}-12}$

(4)

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2. Question 9 · part a · 3 marks · Working and answer

Working and answer

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3. Question 6 · part b · 3 marks

2023 JAN · Paper 01

Further practice · Attempt part b only.

6 Given that $(8 + 3x)^{\frac{1}{3}}$ can be expressed in the form $p(1 + qx)^{\frac{1}{3}}$ where p and q are constants,

(a) find the value of p and the value of q

(2)

(b) Hence, expand $(8 + 3x)^{\frac{1}{3}}$ in ascending powers of x up to and including the term in x^2, expressing each coefficient as an exact fraction in its lowest terms.

(3)

Using the expansion found in part (b) with a suitable value of x

(c) show that $\sqrt[3]{9} \approx \frac{599}{288}$

(2)

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3. Question 6 · part b · 3 marks · Working and answer

Working and answer

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4. Question 6 · part b · 3 marks

2025 JUN · Paper 01

Further practice · Attempt part b only.

6 (a) Show that $\sqrt{4-x} = A\left(1-\frac{x}{B}\right)^{\frac{1}{2}}$ where A and B are integers to be found.

(2)

(b) Hence expand $\sqrt{4-x}$ in ascending powers of x up to and including the term in x^3
Give each coefficient as an exact fraction in its lowest terms.

(3)

(c) Use your expansion with a suitable value of x to obtain an estimate of $\frac{\sqrt{305}}{10}$

Give your answer correct to 5 decimal places.

(3)

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4. Question 6 · part b · 3 marks · Working and answer

Working and answer

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5. Question 6 · part b · 3 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

6 Given that $\sqrt{9-x}$ can be expressed in the form $p(1+qx)^{\frac{1}{2}}$ where p and q are constants

(a) find the value of p and the value of q.

(2)

(b) Hence expand $\sqrt{9-x}$ in ascending powers of x up to and including the term in x^3 expressing each coefficient as an exact fraction in its lowest terms.

(3)

Using the expansion you found in part (b) with a suitable value of x,

(c) find an estimate to 5 decimal places for the value of $\sqrt{\frac{31}{4}}$

(3)

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5. Question 6 · part b · 3 marks · Working and answer

Working and answer

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6. Question 1 · part a · 3 marks

2023 JAN · Paper 02 (regional R)

Further practice · Attempt part a only.

1 (a) Expand $\left(1+\frac{x}{4}\right)^8$ in ascending powers of x up to and including the term in x^3

Give each coefficient in its simplest terms.

(3)

(b) Use your expansion with a suitable value of x to obtain an approximation, to 4 decimal places, of $(1.035)^8$

(3)

(Total for Question 1 is 6 marks)

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6. Question 1 · part a · 3 marks · Working and answer

Working and answer

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7. Question 3 · part a · 3 marks

2020 NOV · Paper 02

Further practice · Attempt part a only.

3 (a) Expand

$$(1 + px)^{-5} \quad p \neq 0$$

in ascending powers of x, up to and including the term in x^4
Give each term in its simplest form.

(3)

The coefficient of x^r in the expansion is c_r

Given that $c_4 = 2c_3$

(b) find the value of p.

(2)

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7. Question 3 · part a · 3 marks · Working and answer

Working and answer

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8. Question 9 · part a · 3 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part a only.

9 (a) Expand $(1+2x)^{-\frac{1}{3}}$ in ascending powers of x up to and including the term in x^3 expressing each coefficient as a fraction in its lowest terms.

(3)

(b) Find the range of values of x for which your expansion is valid.

(1)

$$f(x) = \frac{2+kx^2}{(1+2x)^{\frac{1}{3}}}$$

(c) Obtain a series expansion of f (x) in ascending powers of x up to and including the term in x^3
Give your coefficients in terms of k where appropriate.

(3)

The coefficient of x^3 in the series expansion of f (x) is $-\frac{8}{3}$

(d) Find the exact value of k

(2)

(e) Hence, using algebraic integration, estimate the value of

$$\int_{0.1}^{0.2} f(x)\,dx$$

Give your answer to 4 decimal places.

(5)

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8. Question 9 · part a · 3 marks · Working and answer

Working and answer

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9. Question 2 · part a · 4 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part a only.

2 (a) Expand $\frac{2}{\sqrt{1+3x}}$ in ascending powers of x up to and including the term in x^3

Express each coefficient as a fraction in its simplest terms where appropriate.

(4)

(b) State the range of values of x for which the expansion is valid.

(1)

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9. Question 2 · part a · 4 marks · Working and answer

Working and answer

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10. Question 1 · part whole · 3 marks

2025 NOV · Paper 01

Further practice · Attempt part whole only.

1 Expand $\left(1-\frac{x}{4}\right)^{-2}$ in ascending powers of x up to and including the term in x^3

Where appropriate, express each coefficient as an exact fraction in its lowest terms.

(3)

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(Total for Question 1 is 3 marks)

10. Question 1 · part whole · 3 marks · Working and answer

Working and answer

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11. Question 5 · part b · 3 marks

2022 JUN · Paper 01

Further practice · Attempt part b only.

5 Given that $(2 + 3x)^{-1}$ can be expressed in the form $p(1 + qx)^{-1}$ where p and q are constants,

(a) find the value of p and the value of q

(2)

(b) Hence expand $(2 + 3x)^{-1}$ in ascending powers of x up to and including the term in x^3, expressing each coefficient as an exact fraction in its lowest terms.

(3)

$$f(x) = \frac{1 + x}{2 + 3x}$$

(c) Obtain a series expansion for $f(x)$, in ascending powers of x up to and including the term in x^3, expressing each coefficient as an exact fraction in its lowest terms.

(2)

(d) Hence use algebraic integration to obtain an estimate, to 4 decimal places, of

$$\int_0^{0.5} f(x)\,dx$$

(4)

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11. Question 5 · part b · 3 marks · Working and answer

Working and answer

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12. Question 7 · part a · 3 marks

2024 JUN · Paper 02

Further practice · Attempt part a only.

7 (a) Expand $\left(1+2x^2\right)^{-\frac{3}{4}}$ in ascending powers of x up to and including the term in x^6

Express each coefficient as an exact fraction in its lowest terms.

(3)

$$f(x) = \frac{(2+kx)}{\left(1+2x^2\right)^{\frac{3}{4}}} \quad \text{where } k \neq 0$$

(b) Obtain a series expansion for $f(x)$ in ascending powers of x up to and including the term in x^5

Give each coefficient in terms of k where appropriate.

(2)

The coefficient of the term in x^5 is fourteen times the coefficient of the term in x^2

(c) Find the value of k

(2)

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12. Question 7 · part a · 3 marks · Working and answer

Working and answer

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13. Question 9 · part b · 3 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part b only.

9 Given that $\frac{1}{(2-x)^3}$ can be written as $p(1-qx)^{-3}$

(a) find the value of p and the value of q.

(2)

(b) Expand $\frac{1}{(2-x)^3}$ in ascending powers of x up to and including the term in x^3 and express each coefficient as an exact fraction in its lowest terms.

(3)

$$f(x) = \frac{a+bx}{(2-x)^3} \text{ where } a \text{ and } b \text{ are integers}$$

The first three terms of the expansion of $f(x)$ are $\frac{3}{8} - \frac{43}{16}x + cx^2$

(c) Find the value of a and the value of b.

(3)

(d) Find the exact value of c.

(2)

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13. Question 9 · part b · 3 marks · Working and answer

Working and answer

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14. Question 9 · part b · 3 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

9 (a) Write $\frac{3}{(3-x)^3}$ in the form $a(1-bx)^{-3}$
where a and b are fractions in their lowest terms.

(2)

(b) Expand $\frac{3}{(3-x)^3}$ in ascending powers of x up to and including the term in x^3
Express each coefficient as a fraction in its lowest terms.

(3)

(c) (i) Use a suitable value of x with your expansion in part (b), to obtain an approximation for $\frac{24}{125}$ to 5 decimal places.

(ii) Find the percentage error, to 2 decimal places, of your approximation from the actual value.

(4)

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14. Question 9 · part b · 3 marks · Working and answer

Working and answer

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15. Question 10 · part b · 3 marks

2021 NOV · Paper 02

Further practice · Attempt part b only.

10 (a) Show that $\frac{3}{\sqrt{9-3x}} = \left(1-\frac{x}{3}\right)^{-\frac{1}{2}}$

(2)

(b) Hence expand $\frac{3}{\sqrt{9-3x}}$ in ascending powers of x up to and including the term in x^3 expressing each coefficient as an exact fraction in its lowest terms.

(3)

$$f(x) = \frac{1+2x}{\sqrt{9-3x}}$$

(c) Find the expansion of $3f(x)$ in ascending powers of x up to and including the term in x^3 expressing each coefficient as an exact fraction in its lowest terms.

(4)

(d) Hence, using algebraic integration, obtain an approximation to 6 significant figures for

$$\int_{0.1}^{0.2} \frac{1+2x}{\sqrt{9-3x}}\,dx$$

(4)

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15. Question 10 · part b · 3 marks · Working and answer

Working and answer

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16. Question 7 · part a · 3 marks

2023 JUN · Paper 02

Further practice · Attempt part a only.

7 (a) Expand $\left(1+\frac{x}{3}\right)^{-3}$ in ascending powers of x up to and including the term in x^3

Where appropriate express each coefficient as an exact fraction in its lowest terms.

(3)

(b) Write down the range of values of x for which your expression is valid.

(1)

(c) Express $(3+x)^{-3}$ in the form $P(1+Qx)^{-3}$ where P and Q are rational numbers whose values should be stated.

(2)

$$f(x)=\frac{(1+4x)}{(3+x)^3}$$

(d) Obtain a series expansion for $f(x)$ in ascending powers of x up to and including the term in x^2

(2)

(e) Hence, using algebraic integration, obtain an estimate of $\int_0^{0.2} f(x)\,dx$

Give your answer to 5 significant figures.

(3)

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16. Question 7 · part a · 3 marks · Working and answer

Working and answer

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17. Question 5 · part a · 3 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part a only.

5 (a) Expand $\sqrt{1-x}$ in ascending powers of x up to and including the term in x^3
Give each coefficient as an exact fraction in its lowest terms.

(3)

(b) Using your expansion with a suitable value of x, obtain an approximation, to 6 decimal places, of $\sqrt{0.92}$

(3)

(c) Hence find an approximation, to 5 decimal places, of $\sqrt{23}$

(2)

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17. Question 5 · part a · 3 marks · Working and answer

Working and answer

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18. Question 9 · part a · 3 marks

2023 NOV · Paper 01

Further practice · Attempt part a only.

9 (a) Expand $\left(1-8x^2\right)^{\frac{1}{2}}$ in ascending powers of x, up to and including the term in x^6 giving each coefficient as an integer.

(3)

$$g(x)=\frac{a+bx}{\sqrt{1-8x^2}} \quad \text{where } a \text{ and } b \text{ are prime numbers}$$

Given that the fourth and fifth terms, in ascending powers of x, in the series expansion of g(x) are $20x^3$ and $48x^4$ respectively,

(b) find the value of a and the value of b

(4)

Using the first five terms, in ascending powers of x, in the series expansion of g(x)

(c) obtain an estimate, to 4 significant figures, of $\int_0^{0.2} g(x)\,dx$

(4)

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18. Question 9 · part a · 3 marks · Working and answer

Working and answer

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Practice workbook · When is a binomial expansion valid?

Stating the range of x for which a binomial series with a rational power converges.

3 question parts: 1 worked-example retry and 2 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-SER-19

1. Question 7 · part b · 1 mark

2023 JUN · Paper 02

Worked-example retry · Attempt part b only.

7 (a) Expand $\left(1+\frac{x}{3}\right)^{-3}$ in ascending powers of x up to and including the term in x^3

Where appropriate express each coefficient as an exact fraction in its lowest terms.

(3)

(b) Write down the range of values of x for which your expression is valid.

(1)

(c) Express $(3+x)^{-3}$ in the form $P(1+Qx)^{-3}$ where P and Q are rational numbers whose values should be stated.

(2)

$$f(x)=\frac{(1+4x)}{(3+x)^3}$$

(d) Obtain a series expansion for $f(x)$ in ascending powers of x up to and including the term in x^2

(2)

(e) Hence, using algebraic integration, obtain an estimate of $\int_0^{0.2} f(x)\,dx$

Give your answer to 5 significant figures.

(3)

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1. Question 7 · part b · 1 mark · Working and answer

Working and answer

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2. Question 9 · part b · 1 mark

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

9 (a) Expand $(1 + 2x)^{-\frac{1}{3}}$ in ascending powers of x up to and including the term in x^3 expressing each coefficient as a fraction in its lowest terms.

(3)

(b) Find the range of values of x for which your expansion is valid.

(1)

$$f(x) = \frac{2 + kx^2}{(1 + 2x)^{\frac{1}{3}}}$$

(c) Obtain a series expansion of f (x) in ascending powers of x up to and including the term in x^3
Give your coefficients in terms of k where appropriate.

(3)

The coefficient of x^3 in the series expansion of f (x) is $-\frac{8}{3}$

(d) Find the exact value of k

(2)

(e) Hence, using algebraic integration, estimate the value of

$$\int_{0.1}^{0.2} f(x)\, dx$$

Give your answer to 4 decimal places.

(5)

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2. Question 9 · part b · 1 mark · Working and answer

Working and answer

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3. Question 2 · part b · 1 mark

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

2 (a) Expand $\frac{2}{\sqrt{1+3x}}$ in ascending powers of x up to and including the term in x^3

Express each coefficient as a fraction in its simplest terms where appropriate.

(4)

(b) State the range of values of x for which the expansion is valid.

(1)

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3. Question 2 · part b · 1 mark

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

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Question 2 continued

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(Total for Question 2 is 5 marks)

3. Question 2 · part b · 1 mark · Working and answer

Working and answer

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Practice workbook · Getting a bracket ready for the binomial series: a(1 + bx)^n

Rewriting a bracket such as (4 + 3x)^(1/2) or (2 - x)^(-3) as a constant times (1 + bx)^n, ready for a binomial expansion.

9 question parts: 1 worked-example retry and 8 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-LOG-02

1. Question 5 · part a · 2 marks

2022 JUN · Paper 01

Worked-example retry · Attempt part a only.

5 Given that $(2 + 3x)^{-1}$ can be expressed in the form $p(1 + qx)^{-1}$ where p and q are constants,

(a) find the value of p and the value of q

(2)

(b) Hence expand $(2 + 3x)^{-1}$ in ascending powers of x up to and including the term in x^3, expressing each coefficient as an exact fraction in its lowest terms.

(3)

$$f(x) = \frac{1 + x}{2 + 3x}$$

(c) Obtain a series expansion for $f(x)$, in ascending powers of x up to and including the term in x^3, expressing each coefficient as an exact fraction in its lowest terms.

(2)

(d) Hence use algebraic integration to obtain an estimate, to 4 decimal places, of

$$\int_0^{0.5} f(x)\,dx$$

(4)

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1. Question 5 · part a · 2 marks · Working and answer

Working and answer

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2. Question 6 · part a · 2 marks

2023 JAN · Paper 01

Further practice · Attempt part a only.

6 Given that $(8 + 3x)^{\frac{1}{3}}$ can be expressed in the form $p(1 + qx)^{\frac{1}{3}}$ where p and q are constants,

(a) find the value of p and the value of q

(2)

(b) Hence, expand $(8 + 3x)^{\frac{1}{3}}$ in ascending powers of x up to and including the term in x^2, expressing each coefficient as an exact fraction in its lowest terms.

(3)

Using the expansion found in part (b) with a suitable value of x

(c) show that $\sqrt[3]{9} \approx \frac{599}{288}$

(2)

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2. Question 6 · part a · 2 marks · Working and answer

Working and answer

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3. Question 6 · part a · 2 marks

2025 JUN · Paper 01

Further practice · Attempt part a only.

6 (a) Show that $\sqrt{4-x} = A\left(1-\frac{x}{B}\right)^{\frac{1}{2}}$ where A and B are integers to be found.

(2)

(b) Hence expand $\sqrt{4-x}$ in ascending powers of x up to and including the term in x^3
Give each coefficient as an exact fraction in its lowest terms.

(3)

(c) Use your expansion with a suitable value of x to obtain an estimate of $\frac{\sqrt{305}}{10}$

Give your answer correct to 5 decimal places.

(3)

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3. Question 6 · part a · 2 marks · Working and answer

Working and answer

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4. Question 9 · part a · 2 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part a only.

9 Given that $\frac{1}{(2-x)^3}$ can be written as $p(1-qx)^{-3}$

(a) find the value of p and the value of q.

(2)

(b) Expand $\frac{1}{(2-x)^3}$ in ascending powers of x up to and including the term in x^3 and express each coefficient as an exact fraction in its lowest terms.

(3)

$$f(x) = \frac{a+bx}{(2-x)^3} \text{ where } a \text{ and } b \text{ are integers}$$

The first three terms of the expansion of $f(x)$ are $\frac{3}{8} - \frac{43}{16}x + cx^2$

(c) Find the value of a and the value of b.

(3)

(d) Find the exact value of c.

(2)

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4. Question 9 · part a · 2 marks · Working and answer

Working and answer

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5. Question 9 · part a · 2 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part a only.

9 (a) Write $\frac{3}{(3-x)^3}$ in the form $a(1-bx)^{-3}$
where a and b are fractions in their lowest terms.

(2)

(b) Expand $\frac{3}{(3-x)^3}$ in ascending powers of x up to and including the term in x^3
Express each coefficient as a fraction in its lowest terms.

(3)

(c) (i) Use a suitable value of x with your expansion in part (b), to obtain an approximation for $\frac{24}{125}$ to 5 decimal places.

(ii) Find the percentage error, to 2 decimal places, of your approximation from the actual value.

(4)

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5. Question 9 · part a · 2 marks · Working and answer

Working and answer

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6. Question 6 · part a · 2 marks

2024 NOV · Paper 01

Further practice · Attempt part a only.

6 (a) Show that

$$\frac{a}{\sqrt{4+bx}} = \frac{a}{2}\left(1+\frac{bx}{4}\right)^{-\frac{1}{2}}$$ where a and b are positive integers.

(2)

The expansion of $\frac{a}{\sqrt{4+bx}}$ in ascending powers of x can be written as

$$P+Qx+Rx^2+Sx^3$$

where P, Q, R and S are rational numbers.

(b) Show that $Q=-\frac{ab}{16}$ and $S=-\frac{5ab^3}{2048}$

and find P and R in terms of a and b, as fractions in their lowest terms.

(4)

Given that $Q=\frac{128}{5}S$ and $R=\frac{9}{256}$

(c) show that $a=3$ and $b=1$

(3)

(d) Hence, using an appropriate value of x, find, to 3 decimal places, an approximate value for $\frac{\sqrt{6}}{2}$

(3)

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6. Question 6 · part a · 2 marks · Working and answer

Working and answer

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7. Question 6 · part a · 2 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part a only.

6 Given that $\sqrt{9-x}$ can be expressed in the form $p(1+qx)^{\frac{1}{2}}$ where p and q are constants

(a) find the value of p and the value of q.

(2)

(b) Hence expand $\sqrt{9-x}$ in ascending powers of x up to and including the term in x^3 expressing each coefficient as an exact fraction in its lowest terms.

(3)

Using the expansion you found in part (b) with a suitable value of x,

(c) find an estimate to 5 decimal places for the value of $\sqrt{\frac{31}{4}}$

(3)

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7. Question 6 · part a · 2 marks · Working and answer

Working and answer

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8. Question 10 · part a · 2 marks

2021 NOV · Paper 02

Further practice · Attempt part a only.

10 (a) Show that $\frac{3}{\sqrt{9-3x}} = \left(1-\frac{x}{3}\right)^{-2}$

(2)

(b) Hence expand $\frac{3}{\sqrt{9-3x}}$ in ascending powers of x up to and including the term in x^3 expressing each coefficient as an exact fraction in its lowest terms.

(3)

$$f(x) = \frac{1+2x}{\sqrt{9-3x}}$$

(c) Find the expansion of $3f(x)$ in ascending powers of x up to and including the term in x^3 expressing each coefficient as an exact fraction in its lowest terms.

(4)

(d) Hence, using algebraic integration, obtain an approximation to 6 significant figures for

$$\int_{0.1}^{0.2} \frac{1+2x}{\sqrt{9-3x}}\,dx$$

(4)

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8. Question 10 · part a · 2 marks · Working and answer

Working and answer

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9. Question 7 · part c · 2 marks

2023 JUN · Paper 02

Further practice · Attempt part c only.

7 (a) Expand $\left(1+\frac{x}{3}\right)^{-3}$ in ascending powers of x up to and including the term in x^3

Where appropriate express each coefficient as an exact fraction in its lowest terms.

(3)

(b) Write down the range of values of x for which your expression is valid.

(1)

(c) Express $(3+x)^{-3}$ in the form $P(1+Qx)^{-3}$ where P and Q are rational numbers whose values should be stated.

(2)

$$f(x)=\frac{(1+4x)}{(3+x)^3}$$

(d) Obtain a series expansion for $f(x)$ in ascending powers of x up to and including the term in x^2

(2)

(e) Hence, using algebraic integration, obtain an estimate of $\int_0^{0.2} f(x)\,dx$

Give your answer to 5 significant figures.

(3)

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9. Question 7 · part c · 2 marks · Working and answer

Working and answer

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Practice workbook · Finding the first term and common difference of an arithmetic series

Using the nth term a + (n - 1)d to set up and solve equations for a and d, or to find a particular term or unknown.

15 question parts: 1 worked-example retry and 14 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-SER-01

1. Question 2 · part a · 5 marks

2024 NOV · Paper 01

Worked-example retry · Attempt part a only.

2 The sum of the fifth, sixth and seventh terms of an arithmetic series A is nine times the sum of the first and second terms.
The third term of A is 12

(a) Find the first term and common difference of A

(5)

The nth term of A is u_n

(b) Find the value of $\sum_{r=15}^{60} u_r$

(4)

The sum to n terms of A is S_n

Given that $2S_n - 5u_n = 10$

(c) find the value of n

(4)

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1. Question 2 · part a · 5 marks · Working and answer

Working and answer

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2. Question 6 · part b · 2 marks

2020 NOV · Paper 01

Further practice · Attempt part b only.

6 An arithmetic series A has first term a and common difference d.

The sum S_n of the first n terms of A is given by $S_n = n(15 + 2n)$

(a) Find the value of a and the value of d.

(4)

(b) Find the 20th term of A.

(2)

Given that $S_{2p} - 2S_p = 1 + S_{(p-1)}$

(c) find the value of p.

(4)

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2. Question 6 · part b · 2 marks · Working and answer

Working and answer

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3. Question 1 · part whole · 4 marks

2023 JAN · Paper 01

Further practice · Attempt part whole only.

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1 The nth term of an arithmetic series is a_n where

$$a_{10} + a_{11} + a_{12} = 129 \quad \text{and} \quad a_{19} + a_{20} + a_{21} = 237$$

Find a_1

(4)

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(Total for Question 1 is 4 marks)

3. Question 1 · part whole · 4 marks · Working and answer

Working and answer

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4. Question 7 · part a.i · 0 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part a.i only.

7 An arithmetic series P has first term a, common difference d and nth term u_n

Given that $u_5 = 4x + 6$ and that $u_8 = 7x + 3$

(a) (i) show that $d = x - 1$

(ii) find the value of a

(4)

Given further that $u_9 = 42$

(b) find the value of x

(2)

The sum of the first n terms of P is S_n

(c) Find the value of n for which $S_{(n+1)} = 12u_n + 18$

(5)

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4. Question 7 · part a.i · 0 marks · Working and answer

Working and answer

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5. Question 7 · part a.ii · 4 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part a.ii only.

7 An arithmetic series P has first term a, common difference d and nth term u_n

Given that $u_5 = 4x + 6$ and that $u_8 = 7x + 3$

(a) (i) show that $d = x - 1$

(ii) find the value of a

(4)

Given further that $u_9 = 42$

(b) find the value of x

(2)

The sum of the first n terms of P is S_n

(c) Find the value of n for which $S_{(n+1)} = 12u_n + 18$

(5)

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5. Question 7 · part a.ii · 4 marks · Working and answer

Working and answer

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6. Question 7 · part b · 2 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part b only.

7 An arithmetic series P has first term a, common difference d and nth term u_n

Given that $u_5 = 4x + 6$ and that $u_8 = 7x + 3$

(a) (i) show that $d = x - 1$

(ii) find the value of a

(4)

Given further that $u_9 = 42$

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The sum of the first n terms of P is S_n

(c) Find the value of n for which $S_{(n+1)} = 12u_n + 18$

(5)

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6. Question 7 · part b · 2 marks · Working and answer

Working and answer

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7. Question 5 · part b · 2 marks

2024 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

5 The sum of the first 10 terms of an arithmetic series A is $36k + 1$ where k is a constant.

The 6th term of A is $4k + 1$

(a) (i) Find an expression in terms of k for the common difference of A

(ii) Show that the first term of A is -8

(5)

Given that the 4th term of A is 7

(b) show that $k = 4$

(2)

The sum of the first n terms of A is S_n and the nth term of A is U_n

(c) Find the value of n such that $S_n = 5U_{n+10} + 105$

(4)

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7. Question 5 · part b · 2 marks · Working and answer

Working and answer

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8. Question 1 · part a.i · 0 marks

2020 JAN · Paper 01

Further practice · Attempt part a.i only.

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1 The nth term of an arithmetic series is t_n and the common difference of the series is d.

Given that $t_2 + t_9 = 0$ and that $t_4 + t_6 + t_{10} = 14$

(a) (i) show that $d = 4$

(ii) find the first term of this series.

(4)

A different arithmetic series A has first term 24 and common difference 6
For series A, the sum of the first $2n$ terms is 3 times the sum of the first n terms.

(b) Find the value of n.

(5)

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8. Question 1 · part a.i · 0 marks · Working and answer

Working and answer

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9. Question 1 · part a.ii · 4 marks

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Further practice · Attempt part a.ii only.

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1 The nth term of an arithmetic series is t_n and the common difference of the series is d.

Given that $t_2 + t_9 = 0$ and that $t_4 + t_6 + t_{10} = 14$

(a) (i) show that $d = 4$

(ii) find the first term of this series.

(4)

A different arithmetic series A has first term 24 and common difference 6
For series A, the sum of the first $2n$ terms is 3 times the sum of the first n terms.

(b) Find the value of n.

(5)

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9. Question 1 · part a.ii · 4 marks · Working and answer

Working and answer

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10. Question 2 · part a.i · 0 marks

2022 JAN · Paper 01 (regional R)

Further practice · Attempt part a.i only.

2 The nth term of an arithmetic series is u_n

Given that $u_5 = 46$ and that $u_{20} = 181$

(a) find

(i) the common difference of this series,

(ii) the first term of this series.

(4)

(b) Evaluate $\sum_{n=21}^{50} u_n$

(4)

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10. Question 2 · part a.i · 0 marks · Working and answer

Working and answer

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11. Question 2 · part a.ii · 4 marks

2022 JAN · Paper 01 (regional R)

Further practice · Attempt part a.ii only.

2 The nth term of an arithmetic series is u_n

Given that $u_5 = 46$ and that $u_{20} = 181$

(a) find

(i) the common difference of this series,

(ii) the first term of this series.

(4)

(b) Evaluate $\sum_{n=21}^{50} u_n$

(4)

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11. Question 2 · part a.ii · 4 marks · Working and answer

Working and answer

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12. Question 8 · part a · 6 marks

2023 NOV · Paper 02

Further practice · Attempt part a only.

8 The sum to n terms of an arithmetic series A is S_n

The sum of the first four terms of A is 42 and the fifth term of A is 23

(a) Show that $S_n = \sum_{r=1}^{n}(Pr - Q)$ where P and Q are prime numbers.

(6)

$S_{2n} - 3U_n = 1062$ where U_n is the nth term of A

(b) Find the value of n

(4)

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12. Question 8 · part a · 6 marks · Working and answer

Working and answer

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13. Question 1 · part whole · 5 marks

2022 JUN · Paper 01 (regional R)

Further practice · Attempt part whole only.

1 An arithmetic series has 5th term 16 and 100th term 301

Find the sum of the first 50 terms of the series.

(5)

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(Total for Question 1 is 5 marks)

13. Question 1 · part whole · 5 marks · Working and answer

Working and answer

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14. Question 5 · part a.i · 0 marks

2024 JUN · Paper 01 (regional R)

Further practice · Attempt part a.i only.

5 The sum of the first 10 terms of an arithmetic series A is $36k + 1$ where k is a constant.

The 6th term of A is $4k + 1$

(a) (i) Find an expression in terms of k for the common difference of A

(ii) Show that the first term of A is -8

(5)

Given that the 4th term of A is 7

(b) show that $k = 4$

(2)

The sum of the first n terms of A is S_n and the nth term of A is U_n

(c) Find the value of n such that $S_n = 5U_{n+10} + 105$

(4)

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14. Question 5 · part a.i · 0 marks · Working and answer

Working and answer

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15. Question 5 · part a.ii · 5 marks

2024 JUN · Paper 01 (regional R)

Further practice · Attempt part a.ii only.

5 The sum of the first 10 terms of an arithmetic series A is $36k + 1$ where k is a constant.

The 6th term of A is $4k + 1$

(a) (i) Find an expression in terms of k for the common difference of A

(ii) Show that the first term of A is -8

(5)

Given that the 4th term of A is 7

(b) show that $k = 4$

(2)

The sum of the first n terms of A is S_n and the nth term of A is U_n

(c) Find the value of n such that $S_n = 5U_{n+10} + 105$

(4)

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15. Question 5 · part a.ii · 5 marks · Working and answer

Working and answer

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Practice workbook · Using a binomial expansion to estimate a value

Choosing a value of x that turns the bracket into the required number and substituting into the truncated series to estimate it.

12 question parts: 1 worked-example retry and 11 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-SER-20

1. Question 9 · part b · 3 marks

2020 JAN · Paper 01

Worked-example retry · Attempt part b only.

9 (a) Expand $\sqrt{1-4x}$ in ascending powers of x up to and including the term in x^3, giving each coefficient as an integer.

(3)

(b) Use your expansion with a suitable value for x to obtain an estimate of $\sqrt{0.76}$
Give your answer correct to 4 decimal places.

(3)

(c) Hence find, to 3 decimal places, an estimate of $\sqrt{19}$

(2)

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1. Question 9 · part b · 3 marks · Working and answer

Working and answer

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2. Question 9 · part d · 4 marks

2022 JAN · Paper 01

Further practice · Attempt part d only.

9 (a) Expand $\frac{1}{\sqrt{1-2x}}$ in ascending powers of x up to and including the term in x^3

Simplify each term as far as possible.

(3)

(b) Show that $\frac{1}{\sqrt{0.96}} = \frac{5\sqrt{6}}{12}$

Show all your working clearly.

(2)

(c) Express $\frac{1}{5\sqrt{6}-12}$ in the form $\frac{a\sqrt{6}}{c} + b$ where a, b and c are integers.

Show all your working clearly.

(2)

Using the expansion you found in part (a) with a suitable value of x

(d) find an estimate, to 5 decimal places, of $\frac{9}{5\sqrt{6}-12}$

(4)

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2. Question 9 · part d · 4 marks

2022 JAN · Paper 01

Further practice · Attempt part d only.

Question 9 continued

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2. Question 9 · part d · 4 marks

2022 JAN · Paper 01

Further practice · Attempt part d only.

Question 9 continued

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2. Question 9 · part d · 4 marks

2022 JAN · Paper 01

Further practice · Attempt part d only.

Question 9 continued

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(Total for Question 9 is 11 marks)

2. Question 9 · part d · 4 marks · Working and answer

Working and answer

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3. Question 6 · part c · 2 marks

2023 JAN · Paper 01

Further practice · Attempt part c only.

6 Given that $(8 + 3x)^{\frac{1}{3}}$ can be expressed in the form $p(1 + qx)^{\frac{1}{3}}$ where p and q are constants,

(a) find the value of p and the value of q

(2)

(b) Hence, expand $(8 + 3x)^{\frac{1}{3}}$ in ascending powers of x up to and including the term in x^2, expressing each coefficient as an exact fraction in its lowest terms.

(3)

Using the expansion found in part (b) with a suitable value of x

(c) show that $\sqrt[3]{9} \approx \frac{599}{288}$

(2)

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3. Question 6 · part c · 2 marks

2023 JAN · Paper 01

Further practice · Attempt part c only.

Question 6 continued

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(Total for Question 6 is 7 marks)

3. Question 6 · part c · 2 marks · Working and answer

Working and answer

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4. Question 6 · part c · 3 marks

2025 JUN · Paper 01

Further practice · Attempt part c only.

6 (a) Show that $\sqrt{4-x} = A\left(1-\frac{x}{B}\right)^{\frac{1}{2}}$ where A and B are integers to be found.

(2)

(b) Hence expand $\sqrt{4-x}$ in ascending powers of x up to and including the term in x^3
Give each coefficient as an exact fraction in its lowest terms.

(3)

(c) Use your expansion with a suitable value of x to obtain an estimate of $\frac{\sqrt{305}}{10}$

Give your answer correct to 5 decimal places.

(3)

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4. Question 6 · part c · 3 marks

2025 JUN · Paper 01

Further practice · Attempt part c only.

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Question 6 continued

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(Total for Question 6 is 8 marks)

4. Question 6 · part c · 3 marks · Working and answer

Working and answer

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5. Question 6 · part c · 3 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.

6 Given that $\sqrt{9-x}$ can be expressed in the form $p(1+qx)^{\frac{1}{2}}$ where p and q are constants

(a) find the value of p and the value of q.

(2)

(b) Hence expand $\sqrt{9-x}$ in ascending powers of x up to and including the term in x^3 expressing each coefficient as an exact fraction in its lowest terms.

(3)

Using the expansion you found in part (b) with a suitable value of x,

(c) find an estimate to 5 decimal places for the value of $\sqrt{\frac{31}{4}}$

(3)

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5. Question 6 · part c · 3 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.

Question 6 continued

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5. Question 6 · part c · 3 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.

Question 6 continued

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5. Question 6 · part c · 3 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.

Question 6 continued

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(Total for Question 6 is 8 marks)

5. Question 6 · part c · 3 marks · Working and answer

Working and answer

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6. Question 1 · part b · 3 marks

2023 JAN · Paper 02 (regional R)

Further practice · Attempt part b only.

1 (a) Expand $\left(1+\frac{x}{4}\right)^8$ in ascending powers of x up to and including the term in x^3

Give each coefficient in its simplest terms.

(3)

(b) Use your expansion with a suitable value of x to obtain an approximation, to 4 decimal places, of $(1.035)^8$

(3)

(Total for Question 1 is 6 marks)

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6. Question 1 · part b · 3 marks · Working and answer

Working and answer

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7. Question 9 · part c · 2 marks

2020 JAN · Paper 01

Further practice · Attempt part c only.

9 (a) Expand $\sqrt{1-4x}$ in ascending powers of x up to and including the term in x^3, giving each coefficient as an integer.

(3)

(b) Use your expansion with a suitable value for x to obtain an estimate of $\sqrt{0.76}$ Give your answer correct to 4 decimal places.

(3)

(c) Hence find, to 3 decimal places, an estimate of $\sqrt{19}$

(2)

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7. Question 9 · part c · 2 marks

2020 JAN · Paper 01

Further practice · Attempt part c only.

Question 9 continued

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7. Question 9 · part c · 2 marks

2020 JAN · Paper 01

Further practice · Attempt part c only.

Question 9 continued

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7. Question 9 · part c · 2 marks

2020 JAN · Paper 01

Further practice · Attempt part c only.

Question 9 continued

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(Total for Question 9 is 8 marks)

7. Question 9 · part c · 2 marks · Working and answer

Working and answer

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8. Question 9 · part c.i · 0 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part c.i only.

9 (a) Write $\frac{3}{(3-x)^3}$ in the form $a(1-bx)^{-3}$
where a and b are fractions in their lowest terms.

(2)

(b) Expand $\frac{3}{(3-x)^3}$ in ascending powers of x up to and including the term in x^3
Express each coefficient as a fraction in its lowest terms.

(3)

(c) (i) Use a suitable value of x with your expansion in part (b), to obtain an approximation for $\frac{24}{125}$ to 5 decimal places.

(ii) Find the percentage error, to 2 decimal places, of your approximation from the actual value.

(4)

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8. Question 9 · part c.i · 0 marks · Working and answer

Working and answer

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9. Question 9 · part c.ii · 4 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part c.ii only.

9 (a) Write $\frac{3}{(3-x)^3}$ in the form $a(1-bx)^{-3}$
where a and b are fractions in their lowest terms.

(2)

(b) Expand $\frac{3}{(3-x)^3}$ in ascending powers of x up to and including the term in x^3
Express each coefficient as a fraction in its lowest terms.

(3)

(c) (i) Use a suitable value of x with your expansion in part (b), to obtain an approximation for $\frac{24}{125}$ to 5 decimal places.

(ii) Find the percentage error, to 2 decimal places, of your approximation from the actual value.

(4)

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9. Question 9 · part c.ii · 4 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part c.ii only.

Question 9 continued

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9. Question 9 · part c.ii · 4 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part c.ii only.

Question 9 continued

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9. Question 9 · part c.ii · 4 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part c.ii only.



Question 9 continued

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(Total for Question 9 is 9 marks)

9. Question 9 · part c.ii · 4 marks · Working and answer

Working and answer

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10. Question 6 · part d · 3 marks

2024 NOV · Paper 01

Further practice · Attempt part d only.

6 (a) Show that

$$\frac{a}{\sqrt{4+bx}} = \frac{a}{2}\left(1+\frac{bx}{4}\right)^{-\frac{1}{2}}$$ where a and b are positive integers.

(2)

The expansion of $\frac{a}{\sqrt{4+bx}}$ in ascending powers of x can be written as

$$P+Qx+Rx^2+Sx^3$$

where P, Q, R and S are rational numbers.

(b) Show that $Q=-\frac{ab}{16}$ and $S=-\frac{5ab^3}{2048}$

and find P and R in terms of a and b, as fractions in their lowest terms.

(4)

Given that $Q=\frac{128}{5}S$ and $R=\frac{9}{256}$

(c) show that $a=3$ and $b=1$

(3)

(d) Hence, using an appropriate value of x, find, to 3 decimal places, an approximate value for $\frac{\sqrt{6}}{2}$

(3)

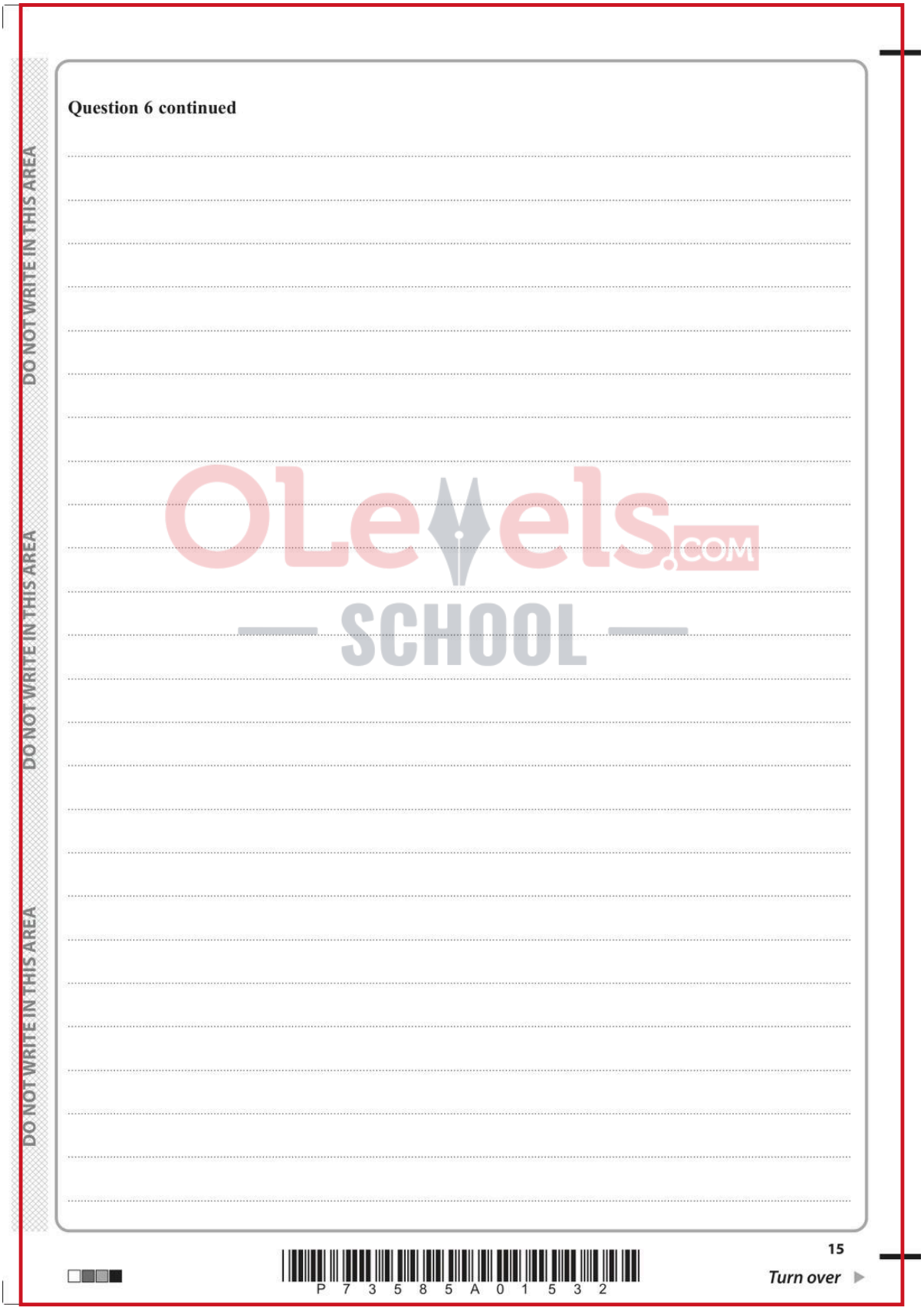
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10. Question 6 · part d · 3 marks

2024 NOV · Paper 01

Further practice · Attempt part d only.



Question 6 continued

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Turn over

10. Question 6 · part d · 3 marks

2024 NOV · Paper 01

Further practice · Attempt part d only.

Question 6 continued

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10. Question 6 · part d · 3 marks

2024 NOV · Paper 01

Further practice · Attempt part d only.

Question 6 continued

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(Total for Question 6 is 12 marks)

10. Question 6 · part d · 3 marks · Working and answer

Working and answer

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11. Question 5 · part b · 3 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part b only.

5 (a) Expand $\sqrt{1-x}$ in ascending powers of x up to and including the term in x^3
Give each coefficient as an exact fraction in its lowest terms.

(3)

(b) Using your expansion with a suitable value of x, obtain an approximation, to 6 decimal places, of $\sqrt{0.92}$

(3)

(c) Hence find an approximation, to 5 decimal places, of $\sqrt{23}$

(2)

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11. Question 5 · part b · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

12. Question 5 · part c · 2 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part c only.

5 (a) Expand $\sqrt{1-x}$ in ascending powers of x up to and including the term in x^3
Give each coefficient as an exact fraction in its lowest terms.

(3)

(b) Using your expansion with a suitable value of x, obtain an approximation, to 6 decimal places, of $\sqrt{0.92}$

(3)

(c) Hence find an approximation, to 5 decimal places, of $\sqrt{23}$

(2)

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12. Question 5 · part c · 2 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part c only.



Question 5 continued

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(Total for Question 5 is 8 marks)

12. Question 5 · part c · 2 marks · Working and answer

Working and answer

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Practice workbook · Finding n from an equation linking sums and terms

Substituting the sum and nth-term formulas into a given equation and solving the quadratic for a positive whole number n.

11 question parts: 1 worked-example retry and 10 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-SER-06

1. Question 1 · part b · 5 marks

2020 JAN · Paper 01

Worked-example retry · Attempt part b only.

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1 The nth term of an arithmetic series is t_n and the common difference of the series is d.

Given that $t_2 + t_9 = 0$ and that $t_4 + t_6 + t_{10} = 14$

(a) (i) show that $d = 4$

(ii) find the first term of this series.

(4)

A different arithmetic series A has first term 24 and common difference 6
For series A, the sum of the first $2n$ terms is 3 times the sum of the first n terms.

(b) Find the value of n.

(5)

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1. Question 1 · part b · 5 marks

2020 JAN · Paper 01

Worked-example retry · Attempt part b only.

Question 1 continued

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1. Question 1 · part b · 5 marks

2020 JAN · Paper 01

Worked-example retry · Attempt part b only.

Question 1 continued

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(Total for Question 1 is 9 marks)

1. Question 1 · part b · 5 marks · Working and answer

Working and answer

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2. Question 6 · part c · 4 marks

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

6 (a) Show that $\sum_{r=1}^{n}(4r-3) = n(2n-1)$

(3)

(b) Hence, or otherwise, find the least value of n such that $\sum_{r=1}^{n}(4r-3) > 1000$

(3)

Given that $S_n = n(2n-1)$, $t_n = (4n-3)$ and that $18 + 3t_{n+7} = S_{n+4}$

(c) find the value of n.

(4)

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2. Question 6 · part c · 4 marks

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 6 continued

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2. Question 6 · part c · 4 marks

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 6 continued

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2. Question 6 · part c · 4 marks

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 6 continued

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(Total for Question 6 is 10 marks)

2. Question 6 · part c · 4 marks · Working and answer

Working and answer

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3. Question 6 · part c · 4 marks

2020 NOV · Paper 01

Further practice · Attempt part c only.

6 An arithmetic series A has first term a and common difference d.

The sum S_n of the first n terms of A is given by $S_n = n(15 + 2n)$

(a) Find the value of a and the value of d.

(4)

(b) Find the 20th term of A.

(2)

Given that $S_{2p} - 2S_p = 1 + S_{(p-1)}$

(c) find the value of p.

(4)

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3. Question 6 · part c · 4 marks

2020 NOV · Paper 01

Further practice · Attempt part c only.



3. Question 6 · part c · 4 marks · Working and answer

Working and answer

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4. Question 8 · part b · 4 marks

2023 NOV · Paper 02

Further practice · Attempt part b only.

8 The sum to n terms of an arithmetic series A is S_n

The sum of the first four terms of A is 42 and the fifth term of A is 23

(a) Show that $S_n = \sum_{r=1}^{n}(Pr - Q)$ where P and Q are prime numbers.

(6)

$S_{2n} - 3U_n = 1062$ where U_n is the nth term of A

(b) Find the value of n

(4)

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4. Question 8 · part b · 4 marks

2023 NOV · Paper 02

Further practice · Attempt part b only.

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Question 8 continued

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(Total for Question 8 is 10 marks)

4. Question 8 · part b · 4 marks · Working and answer

Working and answer

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5. Question 6 · part c · 5 marks

2022 JAN · Paper 02

Further practice · Attempt part c only.

6 The sum of the first n terms of an arithmetic series is S_n where $S_n = n(3 + 2n)$

(a) Find the value of S_{20}

(2)

Given that $S_n = \sum_{r=1}^{n} (Ar + B)$

(b) find the value of A and the value of B

(6)

A different arithmetic series has first term 7 and common difference 4

The sum of the first n terms of this series is T_n

(c) Use algebra to find the value of n for which $T_n = S_n + 252$

(5)

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5. Question 6 · part c · 5 marks

2022 JAN · Paper 02

Further practice · Attempt part c only.

Question 6 continued

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5. Question 6 · part c · 5 marks

2022 JAN · Paper 02

Further practice · Attempt part c only.

Question 6 continued

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5. Question 6 · part c · 5 marks

2022 JAN · Paper 02

Further practice · Attempt part c only.

Question 6 continued

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(Total for Question 6 is 13 marks)

5. Question 6 · part c · 5 marks · Working and answer

Working and answer

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6. Question 7 · part c · 5 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part c only.

7 An arithmetic series P has first term a, common difference d and nth term u_n

Given that $u_5 = 4x + 6$ and that $u_8 = 7x + 3$

(a) (i) show that $d = x - 1$

(ii) find the value of a

(4)

Given further that $u_9 = 42$

(b) find the value of x

(2)

The sum of the first n terms of P is S_n

(c) Find the value of n for which $S_{(n+1)} = 12u_n + 18$

(5)

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6. Question 7 · part c · 5 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 7 continued

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6. Question 7 · part c · 5 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 7 continued

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6. Question 7 · part c · 5 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part c only.



Question 7 continued

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(Total for Question 7 is 11 marks)

6. Question 7 · part c · 5 marks · Working and answer

Working and answer

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7. Question 9 · part c · 5 marks

2021 NOV · Paper 01

Further practice · Attempt part c only.

9 (a) Show that $\sum_{r=1}^{n}(5r-1)=\frac{n}{2}(3+5n)$ (3)

(b) Hence, or otherwise, evaluate $\sum_{r=10}^{20}(5r-1)$ (3)

The sum of the first n terms of an arithmetic series is S_n where $S_n=\sum_{r=1}^{n}(5r-1)$

The rth term of this series is u_r

Given that $S_n=12u_{n+1}+52$

(c) find the value of n (5)

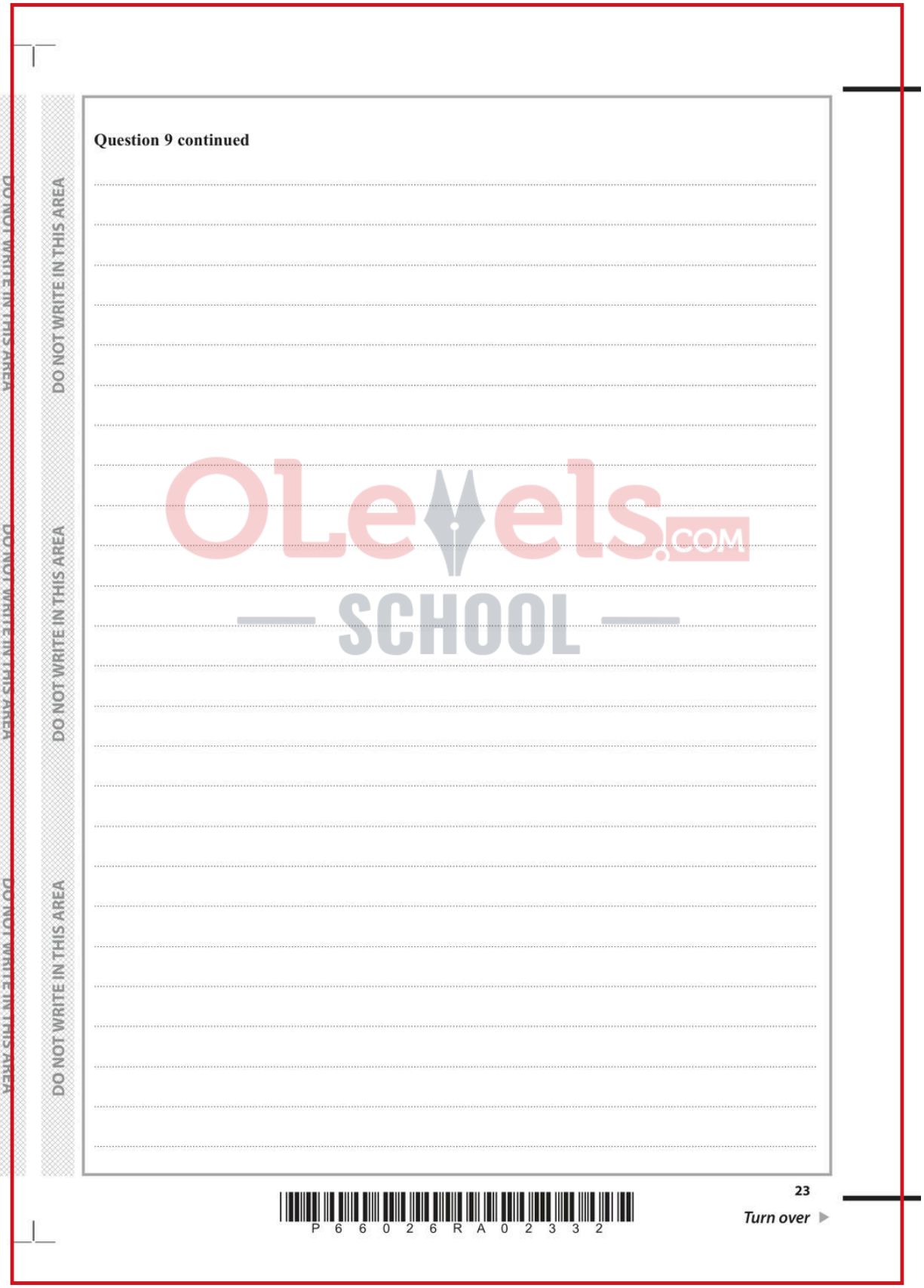
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7. Question 9 · part c · 5 marks

2021 NOV · Paper 01

Further practice · Attempt part c only.



Question 9 continued

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7. Question 9 · part c · 5 marks

2021 NOV · Paper 01

Further practice · Attempt part c only.

Question 9 continued

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7. Question 9 · part c · 5 marks

2021 NOV · Paper 01

Further practice · Attempt part c only.

Question 9 continued

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(Total for Question 9 is 11 marks)

7. Question 9 · part c · 5 marks · Working and answer

Working and answer

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8. Question 2 · part c · 4 marks

2024 NOV · Paper 01

Further practice · Attempt part c only.

2 The sum of the fifth, sixth and seventh terms of an arithmetic series A is nine times the sum of the first and second terms.
The third term of A is 12

(a) Find the first term and common difference of A

(5)

The nth term of A is u_n

(b) Find the value of $\sum_{r=15}^{60} u_r$

(4)

The sum to n terms of A is S_n

Given that $2S_n - 5u_n = 10$

(c) find the value of n

(4)

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8. Question 2 · part c · 4 marks

2024 NOV · Paper 01

Further practice · Attempt part c only.

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Question 2 continued

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(Total for Question 2 is 13 marks)

8. Question 2 · part c · 4 marks · Working and answer

Working and answer

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9. Question 5 · part c · 3 marks

2020 NOV · Paper 02

Further practice · Attempt part c only.

5 (a) Show that $\sum_{r=1}^{n}(3r+5) = \frac{1}{2}n(3n+13)$ (3)

(b) Hence evaluate $\sum_{r=35}^{50}(3r+5)$ (2)

Given that $\sum_{r=1}^{n}(3r+5) = 385$

(c) find the value of n. (3)

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9. Question 5 · part c · 3 marks

2020 NOV · Paper 02

Further practice · Attempt part c only.

Question 5 continued

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(Total for Question 5 is 8 marks)

9. Question 5 · part c · 3 marks · Working and answer

Working and answer

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10. Question 3 · part c · 3 marks

2024 JUN · Paper 01

Further practice · Attempt part c only.

3 (a) Show that $\sum_{r=1}^{n}(5r-3)=\frac{n}{2}(5n-1)$ (3)

(b) Hence, or otherwise, evaluate $\sum_{r=31}^{60}(5r-3)$ (2)

Given that $\sum_{r=1}^{n}(5r-3)=3783$

(c) find the value of n (3)

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10. Question 3 · part c · 3 marks

2024 JUN · Paper 01

Further practice · Attempt part c only.

Question 3 continued

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(Total for Question 3 is 8 marks)

10. Question 3 · part c · 3 marks · Working and answer

Working and answer

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11. Question 5 · part c · 4 marks

2024 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

5 The sum of the first 10 terms of an arithmetic series A is $36k + 1$ where k is a constant.

The 6th term of A is $4k + 1$

(a) (i) Find an expression in terms of k for the common difference of A

(ii) Show that the first term of A is -8

(5)

Given that the 4th term of A is 7

(b) show that $k = 4$

(2)

The sum of the first n terms of A is S_n and the nth term of A is U_n

(c) Find the value of n such that $S_n = 5U_{n+10} + 105$

(4)

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11. Question 5 · part c · 4 marks

2024 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 5 continued

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11. Question 5 · part c · 4 marks

2024 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 5 continued

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12

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11. Question 5 · part c · 4 marks

2024 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

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Question 5 continued

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(Total for Question 5 is 11 marks)

11. Question 5 · part c · 4 marks · Working and answer

Working and answer

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Practice workbook · Finding the first term and common ratio of a geometric series

Writing given terms as ar^(n-1), dividing to remove a, and solving for r and a, or finding a term from the nth-term formula.

17 question parts: 1 worked-example retry and 16 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-SER-08

1. Question 4 · part a.ii · 5 marks

2025 JUN · Paper 02

Worked-example retry · Attempt part a.ii only.

4 A geometric series G has first term a and common ratio r

The third term of G is 10 and the seventh term of G is 40

Given that the second term of G is negative,

(a) find

(i) the exact value of r

(ii) the value of a

(5)

(b) Explain why G does not have a sum to infinity.

(1)

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1. Question 4 · part a.ii · 5 marks · Working and answer

Working and answer

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2. Question 3 · part a.i · 0 marks

2022 JAN · Paper 01

Further practice · Attempt part a.i only.

3 A geometric series G has first term a and common ratio r

The 2nd term of G is $\frac{5}{16}$ and the 5th term of G is $\frac{135}{1024}$

(a) Find

(i) the value of r

(ii) the value of a

(5)

Given that G is convergent with sum to infinity S

(b) find the exact value of S

(2)

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2. Question 3 · part a.i · 0 marks · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 3 · part a.ii · 5 marks

2022 JAN · Paper 01

Further practice · Attempt part a.ii only.

3 A geometric series G has first term a and common ratio r

The 2nd term of G is $\frac{5}{16}$ and the 5th term of G is $\frac{135}{1024}$

(a) Find

(i) the value of r

(ii) the value of a **(5)**

Given that G is convergent with sum to infinity S

(b) find the exact value of S **(2)**

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3. Question 3 · part a.ii · 5 marks · Working and answer

Working and answer

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4. Question 7 · part a.i · 0 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part a.i only.

7 The nth term of a geometric series G is u_n

The first term of G is a and the common ratio of G is r, where $r > 0$

Given that $u_3 = 4$ and that $u_7 = 16$

(a) (i) show that $r = \sqrt{2}$

(ii) find the value of a.

(3)

(b) Find the least value of n for which $u_n > 500$

(4)

The sum of the first n terms of G is S_n

(c) Find S_{20}

Give your answer in the form $p\left(1+\sqrt{2}\right)$ where p is an integer.

(4)

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4. Question 7 · part a.i · 0 marks · Working and answer

Working and answer

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5. Question 7 · part a.ii · 3 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part a.ii only.

7 The nth term of a geometric series G is u_n

The first term of G is a and the common ratio of G is r, where $r > 0$

Given that $u_3 = 4$ and that $u_7 = 16$

(a) (i) show that $r = \sqrt{2}$

(ii) find the value of a.

(3)

(b) Find the least value of n for which $u_n > 500$

(4)

The sum of the first n terms of G is S_n

(c) Find S_{20}

Give your answer in the form $p\left(1+\sqrt{2}\right)$ where p is an integer.

(4)

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5. Question 7 · part a.ii · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

6. Question 5 · part b · 3 marks

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part b only.

5 A geometric series G has first term 12 and common ratio $\frac{3}{8}$

(a) Find the sum to infinity of G

(2)

(b) Show that the 6th term of G can be written as $\frac{3^6}{2^{13}}$

(3)

The nth term of G is u_n

(c) By finding an expression for u_n in terms of n, show that

$$\log_2 u_n = n\log_2 3 - 3n + 5$$

(5)

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6. Question 5 · part b · 3 marks · Working and answer

Working and answer

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7. Question 4 · part a.i · 0 marks

2025 JUN · Paper 02

Further practice · Attempt part a.i only.

4 A geometric series G has first term a and common ratio r

The third term of G is 10 and the seventh term of G is 40

Given that the second term of G is negative,

(a) find

(i) the exact value of r

(ii) the value of a

(5)

(b) Explain why G does not have a sum to infinity.

(1)

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7. Question 4 · part a.i · 0 marks · Working and answer

Working and answer

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8. Question 7 · part a · 3 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part a only.

7 The geometric series G has first term a, common ratio r and nth term u_n

Given that $u_4 = e^{x+2}$ and that $u_7 = e^{\frac{2x+1}{2}}$

(a) show that $r = e^{-\frac{1}{2}}$

(3)

(b) Hence find a in terms of e and x.

(3)

Given that the sum to infinity of G can be written as $\frac{e^p}{e^{\frac{1}{2}} - 1}$

(c) find an expression for p in terms of x.

(3)

Given that $u_{18} > 1.6$ and that x is an integer,

(d) find the least value of x.

(4)

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8. Question 7 · part a · 3 marks · Working and answer

Working and answer

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9. Question 7 · part b · 3 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part b only.

7 The geometric series G has first term a, common ratio r and nth term u_n

Given that $u_4 = e^{x+2}$ and that $u_7 = e^{\frac{2x+1}{2}}$

(a) show that $r = e^{-\frac{1}{2}}$

(3)

(b) Hence find a in terms of e and x.

(3)

Given that the sum to infinity of G can be written as $\dfrac{e^p}{e^{\frac{1}{2}} - 1}$

(c) find an expression for p in terms of x.

(3)

Given that $u_{18} > 1.6$ and that x is an integer,

(d) find the least value of x.

(4)

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P 6 0 4 7 6 A 0 2 0 4 0

9. Question 7 · part b · 3 marks · Working and answer

Working and answer

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10. Question 3 · part a.i · 0 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part a.i only.

3 A geometric series has first term a and common ratio r, where $r > 0$

Given that the 3rd term of the series is 5 and that the 5th term of the series is $\frac{5}{2}$

(a) find

(i) the exact value of r

(ii) the value of a

(4)

(b) Find the sum to infinity of this series.
Give your answer in the form $p + q\sqrt{2}$ where p and q are integers.

(2)

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10. Question 3 · part a.i · 0 marks · Working and answer

Working and answer

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11. Question 3 · part a.ii · 4 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part a.ii only.

3 A geometric series has first term a and common ratio r, where $r > 0$

Given that the 3rd term of the series is 5 and that the 5th term of the series is $\frac{5}{2}$

(a) find

(i) the exact value of r

(ii) the value of a

(4)

(b) Find the sum to infinity of this series.
Give your answer in the form $p + q\sqrt{2}$ where p and q are integers.

(2)

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11. Question 3 · part a.ii · 4 marks · Working and answer

Working and answer

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12. Question 7 · part a · 3 marks

2020 JAN · Paper 02

Further practice · Attempt part a only.

7 The 7th term of a geometric series is 192 and the 8th term of this geometric series is 1152

(a) Find, as a fraction in its simplest form, the 4th term of this geometric series.

(3)

A different geometric series G has a common ratio r and nth term t_n

Given that $t_3 = 24$ and $t_2 + t_3 + t_4 = -36$

(b) show that r satisfies the equation

$$2r^2 + 5r + 2 = 0$$

(5)

Given further that G is convergent with sum to infinity S,

(c) find the value of S.

(4)

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P 5 9 9 3 9 A 0 2 0 3 6

12. Question 7 · part a · 3 marks · Working and answer

Working and answer

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13. Question 8 · part a · 1 mark

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part a only.

8 The nth term of a geometric series G is U_n and the sum of the first n terms of G is S_n

Given that $U_n = \frac{25}{4}\left(\frac{3}{5}\right)^n$

(a) find the exact value of U_5

(1)

(b) Show that $S_n = \sum_{r=1}^{n} \frac{A}{B}\left(\frac{3}{5}\right)^{r-1}$ where A and B are integers to be found.

(3)

The sum to infinity of G is S

(c) Find the least value of n such that $S - S_n < 0.045$

(6)

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13. Question 8 · part a · 1 mark · Working and answer

Working and answer

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14. Question 6 · part a · 2 marks

2025 NOV · Paper 02

Further practice · Attempt part a only.

6 (i) A geometric series G has first term $\left(3+\sqrt{3}\right)$ and common ratio r

The second term of G is $\sqrt{3}$

(a) Find, showing all your working, the exact value of r

Give your answer in the form $\frac{p+\sqrt{3}}{q}$ where p and q are integers to be found.

(2)

(b) Explain why G is convergent.

(1)

(ii) A different geometric series H has first term 8 and common ratio 0.6

The sum to n terms of H is S_n and the sum to infinity of H is T

Find, using logarithms, the least value of n such that $T - S_n < 0.12$

(5)

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14. Question 6 · part a · 2 marks · Working and answer

Working and answer

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15. Question 1 · part whole · 6 marks

2020 NOV · Paper 02 (regional R)

Further practice · Attempt part whole only.

1 The nth term of an arithmetic series A is a_n

The nth term of a geometric series G is t_n

For these two series

$$a_1 = t_1 \qquad a_{10} = t_3 = 48 \qquad a_{10} = 4t_2$$

Find

(i) the common ratio of G,

(ii) the common difference of A.

(6)

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(Total for Question 1 is 6 marks)

15. Question 1 · part whole · 6 marks · Working and answer

Working and answer

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16. Question 7 · part a · 5 marks

2020 NOV · Paper 02

Further practice · Attempt part a only.

7 A geometric series has first term $(x - 3)$, second term $(x + 1)$ and third term $(4x - 2)$.

(a) Find the two possible values of x.

(5)

Given that $x < 1$,

(b) show that the series is convergent.

(2)

The sum to infinity of the series is S.

(c) Find the value of S.

(2)

The sum of the first n terms of the series is S_n

Given that $\frac{S}{S_n} = \frac{256}{255}$

(d) find the value of n.

(3)

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16. Question 7 · part a · 5 marks · Working and answer

Working and answer

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17. Question 6 · part a · 5 marks

2023 JAN · Paper 02

Further practice · Attempt part a only.

6 The nth term of a geometric series G is U_n

The first three terms of G are given by

$$U_1 = q(4p + 1) \qquad U_2 = q(2p + 3) \qquad U_3 = q(2p - 3)$$

(a) Find the possible values of p

(5)

Given that G is convergent with sum to infinity 250

(b) find the value of q

(3)

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P 7 1 8 1 9 A 0 1 6 3 6

17. Question 6 · part a · 5 marks · Working and answer

Working and answer

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Practice workbook · Does a geometric series have a sum to infinity?

Stating or showing that a geometric series has a sum to infinity because $|r| < 1$, or has none because $|r| > 1$.

4 question parts: 1 worked-example retry and 3 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-SER-12

1. Question 7 · part b · 2 marks

2020 NOV · Paper 02

Worked-example retry · Attempt part b only.

7 A geometric series has first term $(x - 3)$, second term $(x + 1)$ and third term $(4x - 2)$.

(a) Find the two possible values of x.

(5)

Given that $x < 1$,

(b) show that the series is convergent.

(2)

The sum to infinity of the series is S.

(c) Find the value of S.

(2)

The sum of the first n terms of the series is S_n

Given that $\frac{S}{S_n} = \frac{256}{255}$

(d) find the value of n.

(3)

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1. Question 7 · part b · 2 marks · Working and answer

Working and answer

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2. Question 4 · part b · 1 mark

2025 JUN · Paper 02

Further practice · Attempt part b only.

4 A geometric series G has first term a and common ratio r

The third term of G is 10 and the seventh term of G is 40

Given that the second term of G is negative,

(a) find

(i) the exact value of r

(ii) the value of a

(5)

(b) Explain why G does not have a sum to infinity.

(1)

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2. Question 4 · part b · 1 mark

2025 JUN · Paper 02

Further practice · Attempt part b only.

Question 4 continued

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(Total for Question 4 is 6 marks)

2. Question 4 · part b · 1 mark · Working and answer

Working and answer

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3. Question 8 · part b · 1 mark

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

8 The sum of the first 2 terms of a geometric series G is 360
The sum of the 2nd and 3rd terms of G is 288

The nth term of G is U_n

(a) Show that $U_n = A\left(\frac{4}{5}\right)^{n-1}$ where A is an integer to be found.

(7)

(b) Explain why G is convergent.

(1)

(c) Hence find the sum to infinity of G

(2)

(d) Find the least number of terms for which the sum is greater than 978

(4)

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3. Question 8 · part b · 1 mark · Working and answer

Working and answer

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4. Question 6 · part b · 1 mark

2025 NOV · Paper 02

Further practice · Attempt part b only.

6 (i) A geometric series G has first term $\left(3+\sqrt{3}\right)$ and common ratio r

The second term of G is $\sqrt{3}$

(a) Find, showing all your working, the exact value of r

Give your answer in the form $\frac{p+\sqrt{3}}{q}$ where p and q are integers to be found.

(2)

(b) Explain why G is convergent.

(1)

(ii) A different geometric series H has first term 8 and common ratio 0.6

The sum to n terms of H is S_n and the sum to infinity of H is T

Find, using logarithms, the least value of n such that $T - S_n < 0.12$

(5)

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4. Question 6 · part b · 1 mark · Working and answer

Working and answer

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Practice workbook · Sum to infinity of a geometric series

Using S = a/(1 - r) once a and r are known or found, sometimes in surd or fraction form.

13 question parts: 1 worked-example retry and 12 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

OLevels.com SCHOOL

Type ID: FPM-SER-13

1. Question 8 · part c · 2 marks

2024 JUN · Paper 01

Worked-example retry · Attempt part c only.

8 The sum of the first and second terms of a geometric series G is 400

The sum of the second and third terms of G is 100

(a) Show that the common ratio of G is $\frac{1}{4}$

(4)

(b) Show that the first term of G is 320

(2)

(c) Find the sum to infinity of G

(2)

The sum to n terms of G is S_n

(d) Find, using logarithms, the least value of n such that

$$S_n > 426.6$$

(4)

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1. Question 8 · part c · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

2. Question 3 · part b · 2 marks

2022 JAN · Paper 01

Further practice · Attempt part b only.

3 A geometric series G has first term a and common ratio r

The 2nd term of G is $\frac{5}{16}$ and the 5th term of G is $\frac{135}{1024}$

(a) Find

(i) the value of r

(ii) the value of a

(5)

Given that G is convergent with sum to infinity S

(b) find the exact value of S

(2)

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2. Question 3 · part b · 2 marks

2022 JAN · Paper 01

Further practice · Attempt part b only.

Question 3 continued

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(Total for Question 3 is 7 marks)

2. Question 3 · part b · 2 marks · Working and answer

Working and answer

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3. Question 5 · part a · 2 marks

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part a only.

5 A geometric series G has first term 12 and common ratio $\frac{3}{8}$

(a) Find the sum to infinity of G

(2)

(b) Show that the 6th term of G can be written as $\frac{3^6}{2^{13}}$

(3)

The nth term of G is u_n

(c) By finding an expression for u_n in terms of n, show that

$$\log_2 u_n = n\log_2 3 - 3n + 5$$

(5)

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3. Question 5 · part a · 2 marks · Working and answer

Working and answer

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4. Question 7 · part c · 2 marks

2020 NOV · Paper 02

Further practice · Attempt part c only.

7 A geometric series has first term $(x - 3)$, second term $(x + 1)$ and third term $(4x - 2)$.

(a) Find the two possible values of x.

(5)

Given that $x < 1$,

(b) show that the series is convergent.

(2)

The sum to infinity of the series is S.

(c) Find the value of S.

(2)

The sum of the first n terms of the series is S_n

Given that $\frac{S}{S_n} = \frac{256}{255}$

(d) find the value of n.

(3)

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4. Question 7 · part c · 2 marks · Working and answer

Working and answer

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5. Question 6 · part b · 3 marks

2023 JAN · Paper 02

Further practice · Attempt part b only.

6 The nth term of a geometric series G is U_n

The first three terms of G are given by

$$U_1 = q(4p + 1) \qquad U_2 = q(2p + 3) \qquad U_3 = q(2p - 3)$$

(a) Find the possible values of p

(5)

Given that G is convergent with sum to infinity 250

(b) find the value of q

(3)

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5. Question 6 · part b · 3 marks

2023 JAN · Paper 02

Further practice · Attempt part b only.

Question 6 continued

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(Total for Question 6 is 8 marks)

5. Question 6 · part b · 3 marks · Working and answer

Working and answer

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6. Question 8 · part c · 2 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.

8 The sum of the first 2 terms of a geometric series G is 360
The sum of the 2nd and 3rd terms of G is 288

The nth term of G is U_n

(a) Show that $U_n = A\left(\frac{4}{5}\right)^{n-1}$ where A is an integer to be found.

(7)

(b) Explain why G is convergent.

(1)

(c) Hence find the sum to infinity of G

(2)

(d) Find the least number of terms for which the sum is greater than 978

(4)

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6. Question 8 · part c · 2 marks · Working and answer

Working and answer

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7. Question 5 · part b · 2 marks

2021 JUN · Paper 02

Further practice · Attempt part b only.

5 The nth term of a geometric series with common ratio r is u_n

Given that $u_2 + u_4 = 212.5$ and that $u_3 + u_4 = 62.5$

(a) find the two possible values of r.

(5)

Given that the series is convergent with sum to infinity S,

(b) find the exact value of S.

(2)

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7. Question 5 · part b · 2 marks

2021 JUN · Paper 02

Further practice · Attempt part b only.



Question 5 continued

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(Total for Question 5 is 7 marks)

7. Question 5 · part b · 2 marks · Working and answer

Working and answer

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8. Question 7 · part c · 3 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part c only.

7 The geometric series G has first term a, common ratio r and nth term u_n

Given that $u_4 = e^{x+2}$ and that $u_7 = e^{\frac{2x+1}{2}}$

(a) show that $r = e^{-\frac{1}{2}}$

(3)

(b) Hence find a in terms of e and x.

(3)

Given that the sum to infinity of G can be written as $\dfrac{e^p}{e^{\frac{1}{2}} - 1}$

(c) find an expression for p in terms of x.

(3)

Given that $u_{18} > 1.6$ and that x is an integer,

(d) find the least value of x.

(4)

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8. Question 7 · part c · 3 marks · Working and answer

Working and answer

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9. Question 3 · part b · 2 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

3 A geometric series has first term a and common ratio r, where $r > 0$

Given that the 3rd term of the series is 5 and that the 5th term of the series is $\frac{5}{2}$

(a) find

(i) the exact value of r

(ii) the value of a

(4)

(b) Find the sum to infinity of this series.
Give your answer in the form $p + q\sqrt{2}$ where p and q are integers.

(2)

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9. Question 3 · part b · 2 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.



Question 3 continued

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(Total for Question 3 is 6 marks)

9. Question 3 · part b · 2 marks · Working and answer

Working and answer

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10. Question 7 · part c · 4 marks

2020 JAN · Paper 02

Further practice · Attempt part c only.

7 The 7th term of a geometric series is 192 and the 8th term of this geometric series is 1152

(a) Find, as a fraction in its simplest form, the 4th term of this geometric series.

(3)

A different geometric series G has a common ratio r and nth term t_n

Given that $t_3 = 24$ and $t_2 + t_3 + t_4 = -36$

(b) show that r satisfies the equation

$$2r^2 + 5r + 2 = 0$$

(5)

Given further that G is convergent with sum to infinity S,

(c) find the value of S.

(4)

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10. Question 7 · part c · 4 marks

2020 JAN · Paper 02

Further practice · Attempt part c only.

Question 7 continued

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10. Question 7 · part c · 4 marks

2020 JAN · Paper 02

Further practice · Attempt part c only.

Question 7 continued

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10. Question 7 · part c · 4 marks

2020 JAN · Paper 02

Further practice · Attempt part c only.

Question 7 continued

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(Total for Question 7 is 12 marks)

10. Question 7 · part c · 4 marks · Working and answer

Working and answer

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11. Question 7 · part b · 2 marks

2023 NOV · Paper 01

Further practice · Attempt part b only.

7 A geometric series G with common ratio r, has first term 16 and third term $\frac{2704}{625}$

(a) Find the two possible values of r

(2)

Given that $r > 0$

(b) find the sum to infinity of G

(2)

The sum to n terms of G is greater than 33

(c) Find, using logarithms, the least possible value of n
Show your working clearly.

(5)

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11. Question 7 · part b · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

12. Question 2 · part whole · 6 marks

2023 JUN · Paper 02

Further practice · Attempt part whole only.

2 The nth term of a convergent geometric series is $8^{(1-2n)}$

Find the sum to infinity of the series.

Give your answer in the form $\frac{p}{q}$ where p and q are integers to be found.

(6)

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12. Question 2 · part whole · 6 marks

2023 JUN · Paper 02

Further practice · Attempt part whole only.

Question 2 continued

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(Total for Question 2 is 6 marks)

12. Question 2 · part whole · 6 marks · Working and answer

Working and answer

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13. Question 4 · part ii · 7 marks

2025 JUN · Paper 02 (regional R)

Further practice · Attempt part ii only.

4 (i) Find the value of $\sum_{k=6}^{17} 3(2)^{k-1}$

(4)

(ii) The common ratio of a geometric series G is positive.

The third term of G is $\frac{7}{13}$ and the ninth term of G is $\frac{448}{9477}$

G is convergent with sum to infinity S

Find the exact value of S

(7)

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13. Question 4 · part ii · 7 marks

2025 JUN · Paper 02 (regional R)

Further practice · Attempt part ii only.

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Question 4 continued

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(Total for Question 4 is 11 marks)

13. Question 4 · part ii · 7 marks · Working and answer

Working and answer

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Practice workbook · Adding up a finite geometric series

Using S_n = a(1 - r^n)/(1 - r) (including for a sigma sum) to find or show a sum, often in exact or surd form.

3 question parts: 1 worked-example retry and 2 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-SER-11

1. Question 7 · part c · 4 marks

2019 JUN · Paper 02 (regional R)

Worked-example retry · Attempt part c only.

7 The nth term of a geometric series G is u_n

The first term of G is a and the common ratio of G is r, where $r > 0$

Given that $u_3 = 4$ and that $u_7 = 16$

(a) (i) show that $r = \sqrt{2}$

(ii) find the value of a.

(3)

(b) Find the least value of n for which $u_n > 500$

(4)

The sum of the first n terms of G is S_n

(c) Find S_{20}

Give your answer in the form $p\left(1+\sqrt{2}\right)$ where p is an integer.

(4)

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1. Question 7 · part c · 4 marks

2019 JUN · Paper 02 (regional R)

Worked-example retry · Attempt part c only.

Question 7 continued

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1. Question 7 · part c · 4 marks

2019 JUN · Paper 02 (regional R)

Worked-example retry · Attempt part c only.

Question 7 continued

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1. Question 7 · part c · 4 marks

2019 JUN · Paper 02 (regional R)

Worked-example retry · Attempt part c only.

Question 7 continued

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(Total for Question 7 is 11 marks)

1. Question 7 · part c · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

2. Question 4 · part i · 4 marks

2025 JUN · Paper 02 (regional R)

Further practice · Attempt part i only.

4 (i) Find the value of $\sum_{k=6}^{17} 3(2)^{k-1}$

(4)

(ii) The common ratio of a geometric series G is positive.

The third term of G is $\frac{7}{13}$ and the ninth term of G is $\frac{448}{9477}$

G is convergent with sum to infinity S

Find the exact value of S

(7)

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2. Question 4 · part i · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 8 · part b · 3 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

8 The nth term of a geometric series G is U_n and the sum of the first n terms of G is S_n

Given that $U_n = \frac{25}{4}\left(\frac{3}{5}\right)^n$

(a) find the exact value of U_5

(1)

(b) Show that $S_n = \sum_{r=1}^{n} \frac{A}{B}\left(\frac{3}{5}\right)^{r-1}$ where A and B are integers to be found.

(3)

The sum to infinity of G is S

(c) Find the least value of n such that $S - S_n < 0.045$

(6)

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3. Question 8 · part b · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

Practice workbook · Finding n in a geometric series with logarithms

Setting up an equation or inequality for a term or sum of a geometric series, isolating r^n and taking logarithms to find n.

14 question parts: 1 worked-example retry and 13 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-SER-14

1. Question 8 · part d · 4 marks

2024 JUN · Paper 01

Worked-example retry · Attempt part d only.

8 The sum of the first and second terms of a geometric series G is 400

The sum of the second and third terms of G is 100

(a) Show that the common ratio of G is $\frac{1}{4}$

(4)

(b) Show that the first term of G is 320

(2)

(c) Find the sum to infinity of G

(2)

The sum to n terms of G is S_n

(d) Find, using logarithms, the least value of n such that

$$S_n > 426.6$$

(4)

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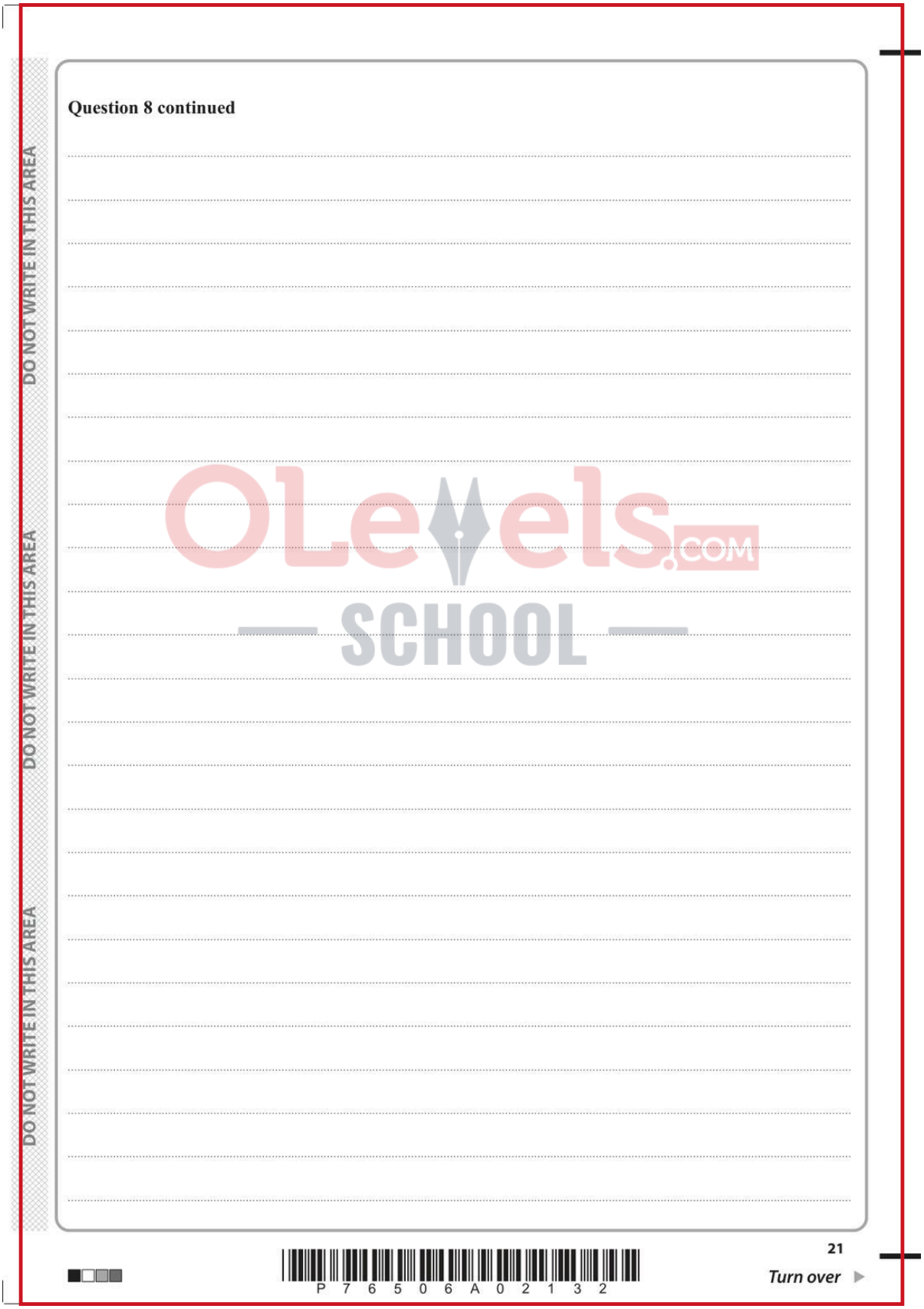
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1. Question 8 · part d · 4 marks

2024 JUN · Paper 01

Worked-example retry · Attempt part d only.



1. Question 8 · part d · 4 marks

2024 JUN · Paper 01

Worked-example retry · Attempt part d only.

Question 8 continued

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1. Question 8 · part d · 4 marks

2024 JUN · Paper 01

Worked-example retry · Attempt part d only.

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Question 8 continued

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(Total for Question 8 is 12 marks)

1. Question 8 · part d · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

2. Question 7 · part b · 4 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

7 The nth term of a geometric series G is u_n

The first term of G is a and the common ratio of G is r, where $r > 0$

Given that $u_3 = 4$ and that $u_7 = 16$

(a) (i) show that $r = \sqrt{2}$

(ii) find the value of a.

(3)

(b) Find the least value of n for which $u_n > 500$

(4)

The sum of the first n terms of G is S_n

(c) Find S_{20}

Give your answer in the form $p\left(1+\sqrt{2}\right)$ where p is an integer.

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2. Question 7 · part b · 4 marks · Working and answer

Working and answer

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3. Question 9 · part b · 3 marks

2023 JAN · Paper 02 (regional R)

Further practice · Attempt part b only.

9 A geometric series G has common ratio r where $r > 0$

The third term of G is $\frac{27}{2}$ and the sum of the first three terms of G is $\frac{57}{2}$

Given that the sum to n terms of G is S_n

(a) show that $S_n = \sum_{j=1}^{n} 4\left(\frac{3}{2}\right)^j$

(8)

Given that $S_k > 50\,000$

(b) show that the least value of k is given by

$$k > \frac{\lg\left(\frac{12\,503}{3}\right)}{\lg\left(\frac{3}{2}\right)}$$

(3)

(c) Hence find the least value of k

(1)

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3. Question 9 · part b · 3 marks · Working and answer

Working and answer

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4. Question 9 · part c · 1 mark

2023 JAN · Paper 02 (regional R)

Further practice · Attempt part c only.

9 A geometric series G has common ratio r where $r > 0$

The third term of G is $\frac{27}{2}$ and the sum of the first three terms of G is $\frac{57}{2}$

Given that the sum to n terms of G is S_n

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(8)

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$$k > \frac{\lg\left(\frac{12\,503}{3}\right)}{\lg\left(\frac{3}{2}\right)}$$

(3)

(c) Hence find the least value of k

(1)

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4. Question 9 · part c · 1 mark

2023 JAN · Paper 02 (regional R)

Further practice · Attempt part c only.

Question 9 continued

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4. Question 9 · part c · 1 mark

2023 JAN · Paper 02 (regional R)

Further practice · Attempt part c only.

Question 9 continued

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4. Question 9 · part c · 1 mark

2023 JAN · Paper 02 (regional R)

Further practice · Attempt part c only.



4. Question 9 · part c · 1 mark · Working and answer

Working and answer

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5. Question 7 · part d · 3 marks

2020 NOV · Paper 02

Further practice · Attempt part d only.

7 A geometric series has first term $(x - 3)$, second term $(x + 1)$ and third term $(4x - 2)$.

(a) Find the two possible values of x.

(5)

Given that $x < 1$,

(b) show that the series is convergent.

(2)

The sum to infinity of the series is S.

(c) Find the value of S.

(2)

The sum of the first n terms of the series is S_n

Given that $\frac{S}{S_n} = \frac{256}{255}$

(d) find the value of n.

(3)

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5. Question 7 · part d · 3 marks

2020 NOV · Paper 02

Further practice · Attempt part d only.

Question 7 continued

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5. Question 7 · part d · 3 marks

2020 NOV · Paper 02

Further practice · Attempt part d only.

Question 7 continued

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5. Question 7 · part d · 3 marks

2020 NOV · Paper 02

Further practice · Attempt part d only.



Question 7 continued

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(Total for Question 7 is 12 marks)

5. Question 7 · part d · 3 marks · Working and answer

Working and answer

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6. Question 8 · part c · 6 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

8 The nth term of a geometric series G is U_n and the sum of the first n terms of G is S_n

Given that $U_n = \frac{25}{4}\left(\frac{3}{5}\right)^n$

(a) find the exact value of U_5

(1)

(b) Show that $S_n = \sum_{r=1}^{n} \frac{A}{B}\left(\frac{3}{5}\right)^{r-1}$ where A and B are integers to be found.

(3)

The sum to infinity of G is S

(c) Find the least value of n such that $S - S_n < 0.045$

(6)

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6. Question 8 · part c · 6 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 8 continued

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6. Question 8 · part c · 6 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 8 continued

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6. Question 8 · part c · 6 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 8 continued

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(Total for Question 8 is 10 marks)

6. Question 8 · part c · 6 marks · Working and answer

Working and answer

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7. Question 8 · part d · 4 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part d only.

8 The sum of the first 2 terms of a geometric series G is 360
The sum of the 2nd and 3rd terms of G is 288

The nth term of G is U_n

(a) Show that $U_n = A\left(\frac{4}{5}\right)^{n-1}$ where A is an integer to be found.

(7)

(b) Explain why G is convergent.

(1)

(c) Hence find the sum to infinity of G

(2)

(d) Find the least number of terms for which the sum is greater than 978

(4)

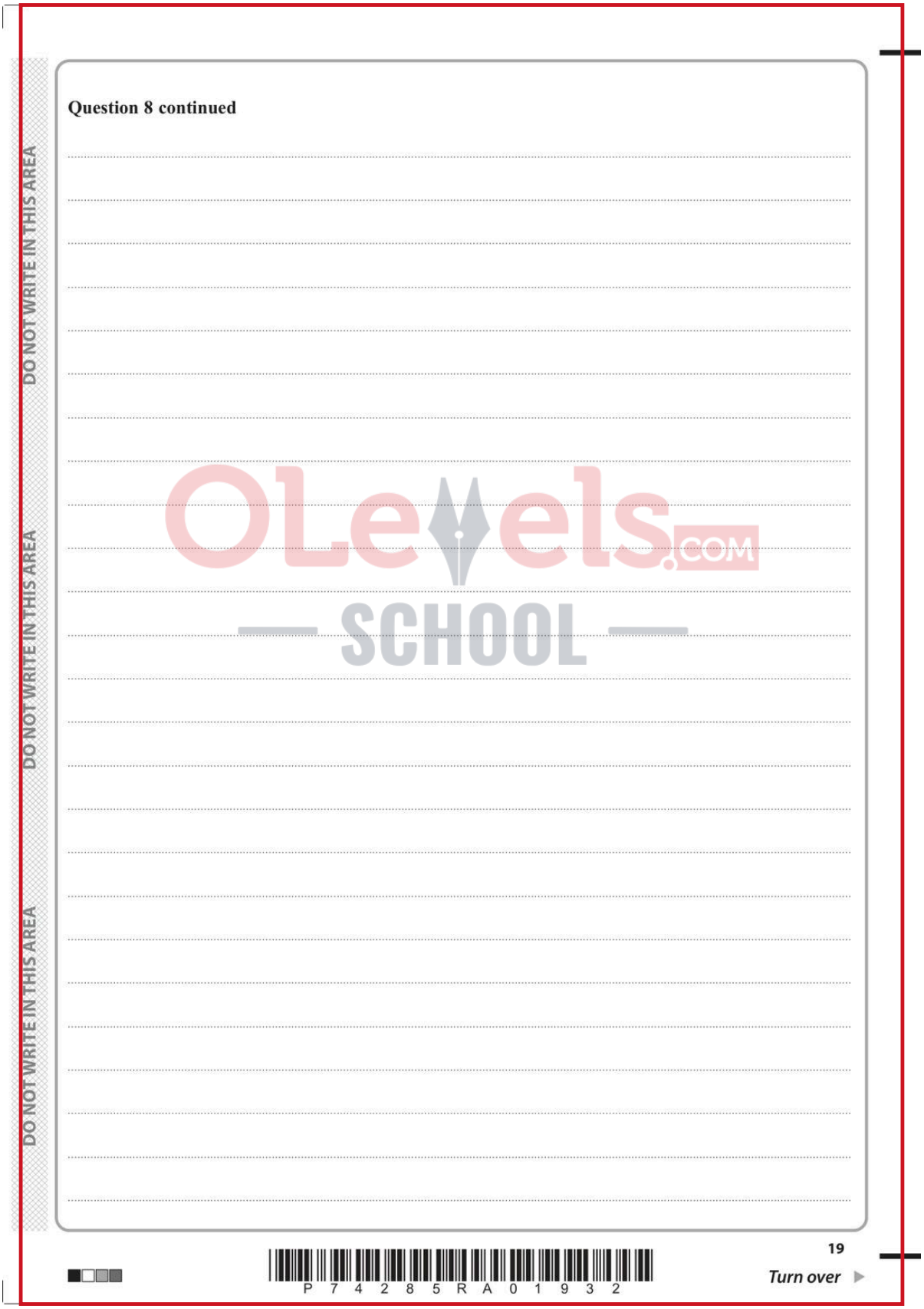
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7. Question 8 · part d · 4 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part d only.



Question 8 continued

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Turn over

7. Question 8 · part d · 4 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part d only.

Question 8 continued

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7. Question 8 · part d · 4 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part d only.

Question 8 continued

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(Total for Question 8 is 14 marks)

7. Question 8 · part d · 4 marks · Working and answer

Working and answer

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8. Question 6 · part b.ii · 5 marks

2025 NOV · Paper 02

Further practice · Attempt part b.ii only.

6 (i) A geometric series G has first term $\left(3+\sqrt{3}\right)$ and common ratio r

The second term of G is $\sqrt{3}$

(a) Find, showing all your working, the exact value of r

Give your answer in the form $\frac{p+\sqrt{3}}{q}$ where p and q are integers to be found.

(2)

(b) Explain why G is convergent.

(1)

(ii) A different geometric series H has first term 8 and common ratio 0.6

The sum to n terms of H is S_n and the sum to infinity of H is T

Find, using logarithms, the least value of n such that $T - S_n < 0.12$

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8. Question 6 · part b.ii · 5 marks

2025 NOV · Paper 02

Further practice · Attempt part b.ii only.

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Question 6 continued

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(Total for Question 6 is 8 marks)

8. Question 6 · part b.ii · 5 marks · Working and answer

Working and answer

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9. Question 7 · part d · 4 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part d only.

7 The geometric series G has first term a, common ratio r and nth term u_n

Given that $u_4 = e^{x+2}$ and that $u_7 = e^{\frac{2x+1}{2}}$

(a) show that $r = e^{-\frac{1}{2}}$

(3)

(b) Hence find a in terms of e and x.

(3)

Given that the sum to infinity of G can be written as $\frac{e^p}{e^{\frac{1}{2}} - 1}$

(c) find an expression for p in terms of x.

(3)

Given that $u_{18} > 1.6$ and that x is an integer,

(d) find the least value of x.

(4)

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9. Question 7 · part d · 4 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part d only.

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Question 7 continued

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9. Question 7 · part d · 4 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part d only.

Question 7 continued

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9. Question 7 · part d · 4 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part d only.



Question 7 continued

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(Total for Question 7 is 13 marks)

9. Question 7 · part d · 4 marks · Working and answer

Working and answer

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10. Question 7 · part b · 2 marks

2021 NOV · Paper 02

Further practice · Attempt part b only.

7 A geometric series G has first term a and common ratio r

The sum of the first three terms of G is $\frac{61}{6}$

The sum to infinity of G is $\frac{125}{6}$

(a) (i) Show that $r = \frac{4}{5}$

(ii) Find the value of a

(6)

The sum of the first n terms of G is S_n

Given that $S_n > 19.8$

(b) show that $n \lg\left(\frac{4}{5}\right) < \lg\left(\frac{31}{625}\right)$

(2)

(c) Hence find the least value of n

(2)

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10. Question 7 · part b · 2 marks · Working and answer

Working and answer

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11. Question 7 · part f · 4 marks

2024 NOV · Paper 02

Further practice · Attempt part f only.

7 (a) Use the factor theorem to show that $(4x-1)$ is a factor of

$$f(x) = 64x^3 - 64x^2 + 3$$

(2)

(b) Hence, or otherwise, find the exact roots of the equation

$$f(x) = 0$$

(4)

A geometric series G has first term a and common ratio r
The third term of G is 9 and the sum to infinity of G is 192

(c) Show that $64r^3 - 64r^2 + 3 = 0$

(3)

Given that r is a rational number

(d) write down the value of r

(1)

(e) show that $a = 144$

(2)

The sum to n terms of G is S_n

(f) Using logarithms, find the least value of n such that $S_n > 191.9$

(4)

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11. Question 7 · part f · 4 marks

2024 NOV · Paper 02

Further practice · Attempt part f only.

Question 7 continued

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11. Question 7 · part f · 4 marks

2024 NOV · Paper 02

Further practice · Attempt part f only.

Question 7 continued

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11. Question 7 · part f · 4 marks

2024 NOV · Paper 02

Further practice · Attempt part f only.

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Question 7 continued

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(Total for Question 7 is 16 marks)

11. Question 7 · part f · 4 marks · Working and answer

Working and answer

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12. Question 7 · part c · 5 marks

2023 NOV · Paper 01

Further practice · Attempt part c only.

7 A geometric series G with common ratio r, has first term 16 and third term $\frac{2704}{625}$

(a) Find the two possible values of r

(2)

Given that $r > 0$

(b) find the sum to infinity of G

(2)

The sum to n terms of G is greater than 33

(c) Find, using logarithms, the least possible value of n
Show your working clearly.

(5)

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12. Question 7 · part c · 5 marks

2023 NOV · Paper 01

Further practice · Attempt part c only.

Question 7 continued

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(Total for Question 7 is 9 marks)

12. Question 7 · part c · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL

13. Question 7 · part c · 5 marks

2019 JUN · Paper 02

Further practice · Attempt part c only.

7 The sum of the first n terms of an arithmetic series is A_n where

$$A_n = \sum_{r=1}^{n} (4r + 5)$$

(a) For this arithmetic series, find

(i) the first term,

(ii) the common difference.

(2)

The sum of the first n terms of a geometric series is G_n where

$$G_n = \sum_{r=1}^{n} 4(3)^{r-1}$$

(b) For this geometric series, find

(i) the first term,

(ii) the common ratio.

(2)

(c) Find the value of n for which $A_{14} - 6 = G_n$

(5)

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13. Question 7 · part c · 5 marks

2019 JUN · Paper 02

Further practice · Attempt part c only.

Question 7 continued

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13. Question 7 · part c · 5 marks

2019 JUN · Paper 02

Further practice · Attempt part c only.

Question 7 continued

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13. Question 7 · part c · 5 marks

2019 JUN · Paper 02

Further practice · Attempt part c only.

Question 7 continued

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(Total for Question 7 is 9 marks)

13. Question 7 · part c · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL

14. Question 7 · part c · 2 marks

2021 NOV · Paper 02

Further practice · Attempt part c only.

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The sum to infinity of G is $\frac{125}{6}$

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(ii) Find the value of a

(6)

The sum of the first n terms of G is S_n

Given that $S_n > 19.8$

(b) show that $n \lg\left(\frac{4}{5}\right) < \lg\left(\frac{31}{625}\right)$

(2)

(c) Hence find the least value of n

(2)

OLevels.com SCHOOL

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14. Question 7 · part c · 2 marks

2021 NOV · Paper 02

Further practice · Attempt part c only.

Question 7 continued

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14. Question 7 · part c · 2 marks

2021 NOV · Paper 02

Further practice · Attempt part c only.

Question 7 continued

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14. Question 7 · part c · 2 marks

2021 NOV · Paper 02

Further practice · Attempt part c only.

Question 7 continued

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(Total for Question 7 is 10 marks)

14. Question 7 · part c · 2 marks · Working and answer

Working and answer

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